

IUSUP ABDULIAPAROV

Unity C# Engineer | Simulation, WebGL and Real-Time 3D Applications

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PROFESSIONAL SUMMARY

I build interactive training and visualization systems with strong attention to architecture, performance and production constraints. Recent work includes a medical angiography simulator with real-time X-ray-style rendering, catheter interaction, procedural workflow logic and a hardware-oriented C-arm control layer. Previously shipped and optimized a high-traffic Unity WebGL product reaching 9M+ users at 40k DAU peak, plus nine standalone WebGL titles built on a shared codebase.

WORK EXPERIENCE

Medical Training Simulator (NDA)

Remote (contract)

Unity Simulation Engineer

2025 to Present

Building core Unity systems for an angiography training simulator covering X-ray-style visualization, guidewire and catheter interaction, contrast injection, C-arm operation and balloon stent procedure logic.

- Worked with clinical reviewers to convert procedure requirements into simulator states, validation rules and instrument profiles.
- Built a **real-time URP X-ray-style rendering pipeline** using density accumulation, multi-pass composition and instrument overlays.
- Implemented **guidewire and catheter interaction**, proximal-end actuation and bifurcation handling, supporting 8 instrument profiles without per-instrument code branches.
- Developed a **procedural balloon/stent workflow** with radius control, vessel deformation, state validation and deployment gating based on positioning and inflation state.
- Designed a **hardware-abstraction layer for C-arm controls**, separating simulator logic from input devices and preparing the system for future physical rig integration.
- Built a **ScriptableObject-driven instrument framework** for collision settings, X-ray layer mapping and reusable rig slots, allowing new instrument profiles to be added through configuration rather than code.

Tech: Unity, C#, URP, ShaderLab/HLSL, Burst, custom render features, ScriptableObjects, procedural meshes.

Aviana Studios, WebGL and Mobile Products

Remote

Unity Engineer | Engineering Owner and Co-founder

2023 to 2026

Owned Unity engineering for a high-traffic Unity WebGL product reaching 9M+ users at 40k DAU peak, plus nine standalone WebGL titles built on a shared codebase.

- Built and maintained a shared Unity codebase shipping nine standalone WebGL titles (Find the Brainrots, Plane Builder, Plus-One Jump, Plus-One Run, Slimes, Build Island and others) from per-game feature branches; reusable services, state machine and save layer across all of them.
- Reduced WebGL build size from 100 MB to 20 MB through shader-variant scoping, asset stripping and texture/audio re-encoding; improved low-end Android from ~15 FPS to stable 60 FPS via GC allocation reduction, pooling, batching and profiler-driven optimization; eliminated mobile-browser page reloads using on-demand Resources.Load with explicit UnloadUnusedAssets / GC.Collect between scene transitions.
- Built a 1k-line custom Rigidbody vehicle controller: spring-damper suspension with bump stops and sphere-cast wheel probes, drift, boost, downforce, air control and auto-recovery; refactored from an earlier ConfigurableJoint prototype to stabilize handling for the game's arcade feel.
- Designed a reusable Unity architecture: DI (VContainer / custom service locator), hierarchical state machines, ScriptableObject configuration and a multi-backend save abstraction (PlayerPrefs / Mirra SDK cloud sync) driving reactive UI through R3 ReactiveProperty.
- Integrated production monetization stack: AppLovin MAX with 13 mediation adapters, Unity IAP, Google Play Billing and Play Core in-app updates, plus five analytics SDKs (GameAnalytics, AppMetrica, AppsFlyer, ByteBrew, Firebase) behind a single facade.
- Optimized environment geometry from ~1M to ~40k triangles on a mobile prototype via mesh decimation, LODs and silhouette-preserving proxies; supported production releases across WebGL and mobile platforms under strict performance, memory and build-size constraints.

Tech: Unity, C#, URP, VContainer, R3, UniTask, Addressables, AppLovin MAX, Mirra SDK, Playgama.

TECHNICAL SKILLS

- **Engine & Languages:** C#, Unity (URP)
- **Architecture:** SOLID, dependency injection (VContainer, custom service locator), hierarchical state machines, ScriptableObject-driven configuration, reactive event systems (R3, UniRx, UniTask), save-system abstraction with multi-backend implementations
- **Rendering:** custom URP render features and passes, image-effect composition, X-ray-style visualization, WebGL shader budgeting, ShaderVariantCollection management, Cinemachine 3, Unity Animation Rigging
- **Physics & Simulation:** custom Rigidbody vehicle controller (spring-damper, sphere-cast, drift/boost/air-control), ConfigurableJoints, guidewire/catheter simulation, multi-rigidbody systems, procedural meshes, blendshape-driven deformation

- **Platform SDKs:** AppLovin MAX (13 mediation adapters), Unity IAP, Google Play Billing & Play Core, Mirra SDK 5, Playgama Bridge (WebGL portals)
- **Analytics:** GameAnalytics, AppMetrica, AppsFlyer, ByteBrew, Firebase Analytics
- **Performance:** Unity Profiler, Memory Profiler, GC allocation reduction, draw-call analysis and batching, WebGL build-size optimization, low-end Android profiling
- **Production Tooling:** Addressables, Unity Localization, custom Editor tools, content pipelines, pre-build validation, Git

EDUCATION

Università Cattolica del Sacro Cuore
BSc Economics & Management
Ufa State Petroleum Tech. University
Automation & Control Systems

Milan, Italy
2024 to Present
Ufa, Russia
2022 to 2024